Real time detection network for multi-scale targets in remote sensing images

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***Abstract*—Object detection in remote sensing imagery is crucial for extracting key information from the vast and growing volume of such data. However, challenges persist due to unique imaging perspectives and wide fields of view, resulting in significant object scale variations and poor detection accuracy for small objects amid complex backgrounds. Although existing methods have made partial progress, limitations remain in enhancing small-object detection and achieving an optimal balance between accuracy and computational efficiency. To address these issues, we propose FSD-YOLO, a novel object detector tailored for remote sensing image analysis, designed to achieve high detection accuracy without compromising efficiency. Our approach introduces two key components: the Feature Complementary Integration Unit (FCU) and the Spatial Mapping Perception Module (SPM), both aimed at improving feature representation and perceptibility of small objects in cluttered scenes. Additionally, we incorporate Dysample to mitigate noise amplification caused by texture artifacts during upsampling. Experimental results on the DIOR dataset show that our method achieves a mAP@0.5 of 75.2%, surpassing the baseline YOLOv8m by 1.8%. Furthermore, FSD-YOLO demonstrates superior performance compared to state-of-the-art real-time detectors, effectively balancing accuracy and speed for practical deployment.**

***Index Terms*—small object detection, remote sensing images, multi-scale feature fusion, context extraction**

I. Introduction

Due to advancements in aerospace and remote sensing technologies, the volume of image data collected from satellites, unmanned aerial vehicles, and aircraft continues to expand significantly, characterized by high resolution and specific overhead perspectives.Object detection in remote sensing images aims to accurately and efficiently identify and localize targets from massive amounts of remote sensing data.Playing a crucial role in enabling automated analysis of remote sensing data and supporting accurate decision-making. It currently suffers from an imbalance between multi-scale object detection accuracy and read speed.

Unlike natural scene images, remote sensing images are captured from an overhead perspective, causing objects to exhibit significant variations in scale and arbitrary orientations. Additionally, small objects are easily overwhelmed by complex background noise. Under conditions of heavy background clutter and large variations in object scales, improving the detection performance of small objects while maintaining high detection speed has remained a key focus in detector research.

The existing mainstream object detection algorithms mainly fall into the following three categories: Two-stage methods, such as the R-CNN series, first generate candidate regions using a Region Proposal Network, and then perform classification and bounding box regression on these regions. Although highly accurate, their performance is limited by high computational cost and time overhead. One-stage methods, such as the YOLO [1] and SSD [2]series based on convolutional neural networks, treat object detection as a direct regression problem without requiring explicit region proposal generation. Transformer-based approaches, such as DETR [3] , utilize an encoder to extract image features and a decoder to generate bounding boxes and class predictions in an end-to-end manner.

Improving the performance of object detection in RSI, researchers have proposed various enhancement strategies. Y. Xiao et al. [4] introduced a space position joint attention mechanism into the YOLOv8 [5] network model, effectively suppressing noise interference caused by complex backgrounds and thereby enhancing the network's feature extraction capability. However, this approach primarily focuses on feature optimization within the spatial dimension and does not adequately account for the differences in noise characteristics across different channels. T. Zhang et al. [6] on the other hand, incorporated a weighted spatial pyramid pooling structure and a weighted bidirectional feature pyramid network during the feature fusion stage, significantly improving the model's detection performance through optimized feature integration.

To tackle the challenges presented by the aforementioned remote sensing image detection tasks, we introduce FSD-YOLO. To evaluate the effectiveness of our approach, we perform comprehensive experiments on the widely-used remote sensing image dataset DIOR. Our key contributions are summarized as follows:

Our goal in designing the FCU is to mitigate semantic information loss for small objects in deep networks. By aligning fine-grained semantics from deeper layers, FCU enhances small object localization. It enables efficient multi-scale feature extraction with minimal parameters and reduces channel-wise redundancy via a transform-and-fuse strategy, thereby enhancing object detection accuracy.

We propose the SPM, which leverages multi-receptive-field convolutions to enhance texture-context relationships across

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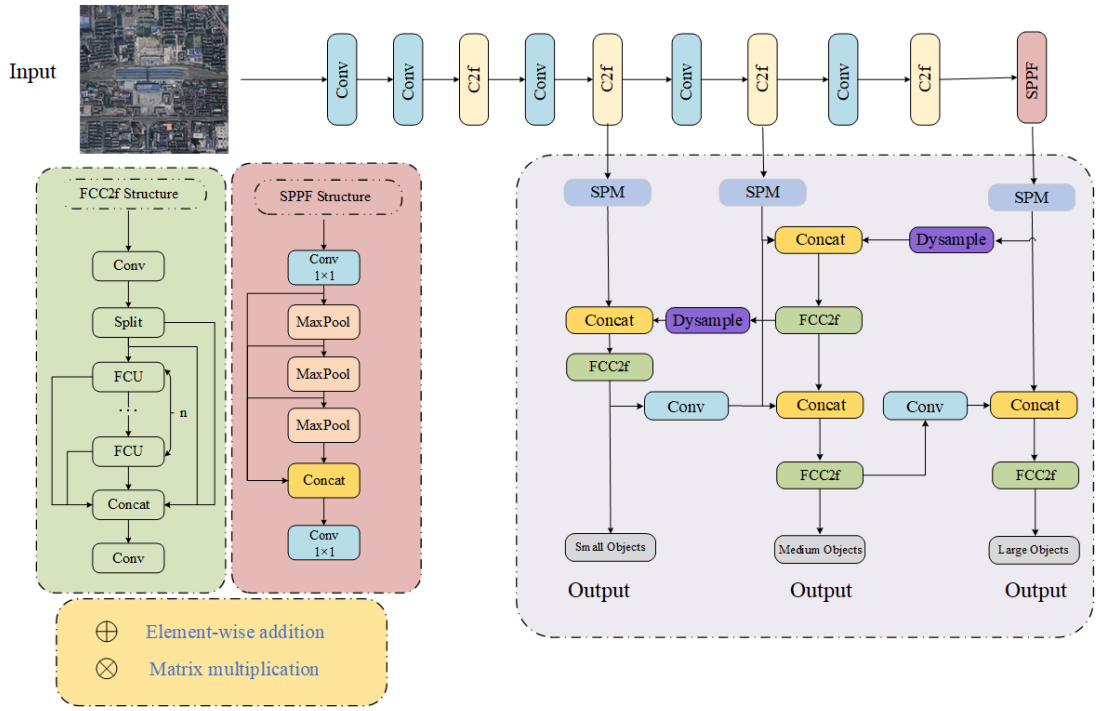


Fig. 1. The Network Structure Of FSD-YOLO

scales, improving sensitivity to objects of varying sizes. By dynamically adjusting spatial and channel-wise feature weights, SPM strengthens multi-scale representation and suppresses background noise.

We introduced DySample [7], a learnable upsampling module that can adaptively resample bilinear interpolation feature maps based on input content. By generating content aware sampling points, it enhances the flexibility of upsampling and reduces the generation of noise textures caused by interpolation.

II. Methodology

This chapter provides a comprehensive analysis of the network architecture of FSD-YOLO and the functionality of its individual modules. We use YOLOv8m as the baseline framework and integrate three specially designed modules into the neck to improve detection performance on remote sensing images. The overall architecture of FSD-YOLO is illustrated in Fig. 1.

A. Feature Complementary Integration Unit

To fully extract and effectively integrate the contextual information and semantic content of multi-scale RSI during the feature fusion process, we propose the FCU module, which consists of three stages: extraction, transformation, and fusion. The overall architecture of FCU is illustrated in Fig. 2.

Extraction: This stage aims to obtain diverse semantic information from the perspective of feature richness. The module design particularly emphasizes the expansion of the receptive field. To this end, we integrate multiple branch structures, each employing convolutional kernels of different sizes, to capture richer local contextual information and multi-scale features. By grouping and summing features within different receptive field ranges and fusing the information, low-scale feature representation X_{low} and high-scale feature representation X_{up} are generated. This strategy effectively promotes collaborative representation and integration of features from different receptive fields across multiple scales. This extraction process can be formally expressed as:

$$X_{up} = \mathrm{DW}_{3\times3}(X) + \mathrm{DW}_{5\times5}(X) \tag{1}$$

$$X_{low} = \mathrm{DW}_{7\times7}(X) + \mathrm{DW}_{9\times9}(X) \tag{2}$$

Transformation: The high-scale feature X_{up} is fed into the upper transformation stage. To efficiently extract high-level semantic representations while reducing computational cost, we employ lightweight convolutional operations—specifically, grouped depthwise separable convolution (GWC) and pointwise convolution (PWC)—as replacements for the computationally expensive standard $K \times K$ convolution. Specifically, GWC significantly decreases model parameter count and computational complexity by employing sparse convolutions across channels. However, its grouping mechanism restricts information exchange between different channel groups, hindering cross-channel feature flow. To address this, a 1×1 pointwise convolution (PWC) is introduced to restore the cross-channel dependencies lost due to grouping, thereby promoting global feature fusion. In implementation, we apply the $K \times K$ GWC and the 1×1 PWC in parallel on the same input x_{up}, and then perform element-wise addition of their outputs to generate the fused high-order feature map Y_1. This upper transformation process can be formally expressed as:

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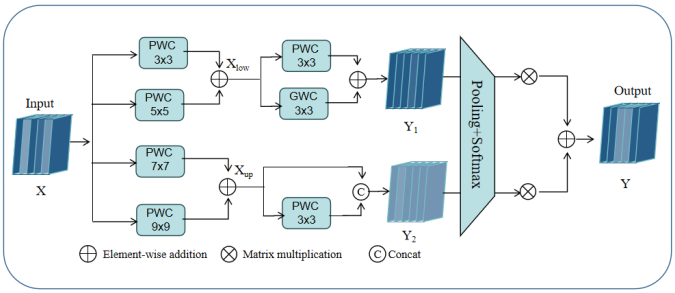


Fig. 2. The Network Structure Of FCU

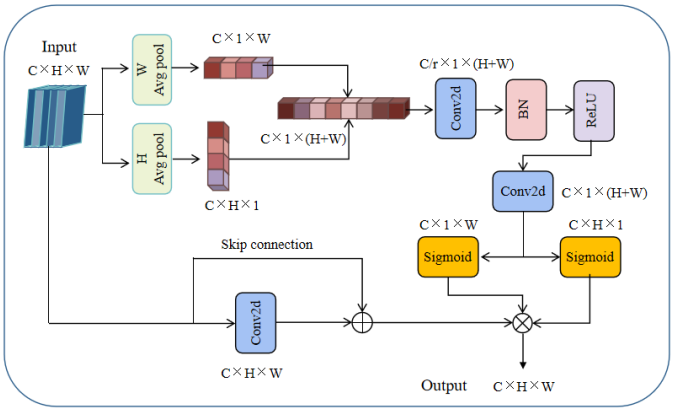


Fig. 3. The Network Structure Of SPM

$$Y_1 = M^{GW} X_{up} + M^{PW1} X_{up} \tag{3}$$

The matrices M^{GW} and M^{PW1} are learnable parameters used in the GWC and PWC operations, respectively. In the upper transformation stage, both GWC and PWC are applied to the same input feature X_{up}, enabling the extraction of a rich and representative feature Y_1 while keeping computational overhead low.

The feature X_{low} is passed into the lower transformation stage. There, a lightweight 1×1 point-wise convolution (PWC) is applied to produce a feature map that captures shallow-level hidden details, which complements the rich feature extraction performed in the upper stage. Furthermore, we directly reuse the original feature X_{low} to generate additional feature maps, without introducing any extra computational burden. These two sets of features—those generated by the PWC and those reused—are then summed element-wise to produce the final output of the lower stage, denoted as Y_2, according to the following expression:

$$Y_2 = M^{PW2} X_{low} \cup X_{low} \tag{4}$$

In summary, the lower transformation stage reuses the previous feature X_{low} and applies a lightweight 1×1 point-wise convolution (PWC) to generate feature Y_2, which contains complementary fine details.

Fusion: Following the transformation, we initially perform global average pooling on the spatial statistical information of the channels S_m, expressed as:

$$P_m = Pooling(Y_m), m = 1, 2 \tag{5}$$

Subsequently, the global channel descriptors from the upper and lower branches, denoted as P_1 and P_2, are concatenated along the channel dimension. A channel-wise soft attention mechanism is then applied to produce the feature importance weights α_1 and α_2, formulated as follows:

$$\alpha_1 = \frac{e^{P_1}}{e^{P_1} + e^{P_2}}, \alpha_2 = \frac{e^{P_2}}{e^{P_1} + e^{P_2}}, \alpha_1 + \alpha_2 = 1 \tag{6}$$

Finally, guided by the feature importance vectors α_1 and α_2, the high-level feature Y_1 and the low-level feature Y_2 are combined channel-wise in the following manner. The fused feature is then processed through a 3×3 convolution to integrate high- and low-dimensional information and adjust the channel dimensions, ultimately yielding the channel-refined feature Y. Its mathematical representation is as follows:

$$Y = \text{Conv}(\alpha_1 Y_1 + \alpha_2 Y_2) \tag{7}$$

B. Spatial Mapping Perception Module

To achieve accurate localization and recognition of small objects in RSI and effectively suppress interference from complex environmental noise, this paper proposes the SPM module. Through this mechanism, the network's ability to jointly perceive global context and local details is enhanced, significantly improving the effectiveness of capturing long-range dependencies and remote contextual information in complex scenes. The overall architecture of FCU is illustrated in Fig. 3.

This module comprises three parallel paths. Two of them form a pair of 1×1 convolutional branches that apply one-dimensional global average pooling along complementary spatial dimensions, respectively, to encode channel-wise features across their corresponding dimensions. The outputs of these two branches are then fed into a fusion convolution, whose result is subsequently split again along the channel dimension into two separate vectors. Each vector is activated by a sigmoid function to generate a spatially directional attention map, effectively modeling a two-dimensional bidirectional attention distribution, thereby enhancing the representational capacity for inter-channel relationships.

The third path employs a standard 3×3 convolution within a residual structure to capture local cross-channel interactions. Finally, the importance weights of each channel are dynamically adjusted through an element-wise multiplication of the outputs from all three branches. This module effectively encodes inter-channel dependencies while preserving fine-grained spatial structural information and embedding it into the channel-wise feature representation, thus achieving collaborative optimization across both spatial and channel dimensions.

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TABLE I
QUANTITATIVE COMPARISONS OF FSD- AND OTHER METHODS ON THE DIOR DATASET

Method	mAP	AL	AT	BF	BC	B	C	D	ESA	ETS	GC	GTF	HB	O	S	SD	ST	TC	TS	V	WM
DETR [3]	68.6	79.3	74.0	65.2	80.7	26.5	75.2	66.8	70.5	52.5	74.2	62.1	27.4	47.0	8.5	46.0	14.5	64.4	54.3	14.4	55.6
MSFC-Net [8]	70.1	85.8	76.2	74.4	91.1	44.2	78.1	55.5	60.9	59.5	76.9	73.7	49.6	57.2	89.6	69.2	76.5	86.7	51.8	55.2	84.3
FSoD-NET [9]	71.8	88.8	66.9	86.8	90.2	45.5	79.6	48.2	**86.9**	**75.5**	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	52.0	**90.6**
DIAG-TR [10]	72.0	68.4	72.2	71.9	85.1	29.8	76.8	60.2	70.3	73.2	75.1	35.5	35.5	50.8	41.6	71.8	32.9	81.0	53.1	33.8	78.6
ABNet [11]	72.8	66.8	83.7	75.2	87.7	**50.3**	78.2	67.8	85.9	74.2	79.7	81.0	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	89.1
SESA-Net [12]	74.6	**93.5**	78.2	**91.9**	82.2	46.8	**89.3**	61.4	67.2	67.5	73.3	76.7	55.8	62.0	78.8	**84.8**	**82.4**	89.8	54.4	**76.7**	79.1
Ours	**75.2**	92.6	**84.0**	75.5	**91.3**	49.6	81.5	**69.3**	71.0	67.8	**80.9**	**81.1**	63.6	**62.7**	**90.4**	68.5	76.0	91.0	**66.8**	56.8	83.2

TABLE II
ABLATION EXPERIMENTD RELATED TO THE DIOR DATASET

Method	mAP	AL	AT	BF	BC	B	C	D	ESA	ETS	GC	GTF	HB	O	S	SD	ST	TC	TS	V	WM
Baseline	73.37	91.43	80,97	73.05	91.76	47.01	82.53	66.81	60.88	61.10	82.85	81.41	66.46	62.35	90.13	63.96	77.59	90.64	59.32	54.38	82.72
+FCU	74.42	91.67	84.66	74.07	91.41	48.79	82.41	66.48	66.38	66.58	81.85	81.48	65.16	62.46	90.31	61.4	75.62	90.77	66.8	56.39	83.74
+SPM	74.41	91.71	81.6	73.12	91.86	49.46	82.7	68.54	63.14	64.49	83.17	81.4	66.17	61.89	89.74	68.74	79.19	91.02	62.41	55.01	82.84
+Dysample	73.94	92.43	82.68	72.12	92.23	48.02	81.98	69.82	63.22	63.1	82.63	80.27	66.25	61.41	89.71	64	78.74	87.13	62.97	57.22	82.88
+SPM+Dysample	75.24	93.32	84.13	75.00	92.12	49.04	82.02	70.19	61.02	61.49	82.28	82.21	67.19	63.87	89.88	72.81	78.94	91.21	68.35	57.13	82.57
+FCU+SPM	75.01	91.9	83.97	76.35	91.91	48.4	81.85	68.66	67.98	66.54	80.88	82.16	65.15	62.47	90.44	65.25	77.35	90.75	67.89	55.93	84.33
+FCU+SPM+Dysample	75.18	92.64	83.97	75.54	91.30	49.60	81.46	69.32	70.99	67.83	80.86	81.14	63.57	62.70	90.44	68.52	75.97	90.98	66.80	56.81	83.20

C. Dynamic Upsampling Module

To minimize the noise introduced during upsampling while maintaining a lightweight design, we introduce the DySample structure, which reformulates the upsampling process using point-wise sampling. The module adopts a parallel branch structure: one simple branch generates the base upsampled feature via bilinear interpolation, while the other branch produces a set of sampling points through linear operations combined with point transposition. These sampling points are then used to resample the input feature, and the resulting feature is added to the output of the bilinear branch to obtain the final upsampled feature map. This structure enables dynamic adjustment of the upsampled features based on contextual information, effectively reducing noise interference during the fusion process, all while having minimal impact on model performance.

III. Experiment

In this section, we will first introduce the DIOR dataset. Then, we will describe the implementation details of the experiments. Finally, we conduct ablation studies, including comparative experiments and result visualizations, to demonstrate the advantages of the FSD-YOLO network.

A. Datasets

The DIOR dataset is a publicly accessible, large-scale dataset for object detection in optical remote sensing images, containing 23,463 images with a total of 190,288 annotated objects. Following the data partitioning strategy described in the original paper, it is split into 11,738 images for training, 5,863 for validation, and 5,862 for testing. The dataset includes 20 distinct object categories: airplane(AL), airport (AT), baseball field (BF), basketball court (BC), bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), golf course (GC), ground track field (GTF),harbor (HB), overpass (O), ship (S), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (V), and windmill (WM).

B. Implement specifications

The experiments were conducted in an environment based on the Linux operating system, implemented using the PyTorch deep learning framework. All training and testing procedures were performed on a workstation equipped with an A800 40GB GPU and 32GB of memory. The optimization algorithm employed was Stochastic Gradient Descent (SGD), configured with an initial learning rate of 0.001 and a momentum value of 0.9. Throughout the training process, the batch size was set to 16 samples per iteration, and all input images were consistently resized to a resolution of 640×640 pixels. To ensure sufficient convergence, we set the number of training epochs for the dataset on the model to 300.

C. Comparison Experiments and Analysis

To verify the effectiveness of the proposed method, we selected several state-of-the-art object detection algorithms and trained and evaluated them on the DIOR dataset, with results shown in Table I. Our model achieved the highest mAP, significantly outperforming the comparative methods, demonstrating its advantages in optical remote sensing image detection. Among the 20 object categories in DIOR, our method performed best in 9 categories. Particularly for small objects such as "ship" and "vehicle", as well as large objects like "ground track field", its average precision (AP) surpassed that of the second-best model. Experiments show that the proposed method possesses stronger multi-scale feature representation and localization capabilities, especially improving detection performance for small objects and effectively enhancing detection results in complex scenarios.

D. Ablation study and analysis

To verify the effectiveness of the proposed network improvements, a series of ablation studies were conducted in this paper. The ablation study consists of seven experimental configurations: Group 1 employs the original YOLOv8m model. The results of the ablation experiments are summarized in Table II.

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Fig. 4. The FSD-YOLO Detection Result

As shown in Table II, in the second experiment group, the FCU module was added to the baseline model, alleviating the imbalance in feature information fusion across different scales. The detection accuracy improved from the original 73.4% to 74.4%, representing a 1.0% increase in mAP. Detection performance for objects of various scales was enhanced to differing degrees. Specifically, the Average Precision (AP) for large-scale objects increased by 3.6% for the "airport" category and by 5.5% for the "expressway service area" category. For small-scale objects, the AP for the "vehicle" class improved by 2.0%.

In the third experimental group, by incorporating the SPM, the mAP on the DIOR dataset increased from 73.4% to 74.4%, representing a 1.1% improvement. For categories prone to interference from complex backgrounds—such as vehicles and ships—the precision of the vehicle class improved from 54.4% to 55.0%, while the precision for the stadium class significantly increased from 64.0% to 68.7%. Experimental results show that the SPM module improves the model's focus on discriminative object features, suppresses background interference, and preserves fine-grained details during feature extraction, especially for small objects.

In the fourth experimental group, by replacing the standard up sampling module with the DySample module, the accuracy on the DIOR dataset increased from 73.4% to 73.9%, achieving a 0.5% improvement, with almost no additional computational cost. The DySample module introduces greater flexibility in the upsampling process and effectively alleviates noise introduced during upsampling.

IV. Conclusion

We propose FSD-YOLO, a framework specifically designed to enhance multi-scale object detection performance in RSI. It can be widely applied in fields such as urban planning, resource management, environmental protection, and military reconnaissance.Its detection results are shown in Fig. 4.

The FCU module effectively extracts multi-scale semantic information and enables the fusion of high and low dimensional features with adaptive channel adjustment. The SPM is designed to strengthen the semantic representation of remote sensing objects and suppress the interference caused by background noise in RSI, thereby improving detection accuracy. Finally, the DySample module focuses on dynamically reconstructing semantic features enhancing the model's robustness in handling complex feature patterns inherent in tasks.

Although the experiments have demonstrated the outstanding performance of FSD-YOLO.However, we observe that under rainy and foggy weather conditions, FSD-YOLO struggles to effectively mitigate the adverse effects of rain and fog, leading to a degradation in detection accuracy. Currently, the method is primarily applicable under relatively clear imaging conditions. In the future, we plan to conduct further in-depth research and incorporate multimodal information to improve the detector's capability to simultaneously identify objects and counteract the interference of rain and fog in remote sensing images, thereby enhancing its adaptability in challenging weather scenarios.

References

[1] M. Hussain, "Yolov1 to v8: Unveiling each variant–a comprehensive review of yolo," *IEEE access*, vol. 12, pp. 42 816–42 833, 2024.

[2] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *European conference on computer vision*. Springer, 2016, pp. 21–37.

[3] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.

[4] Y. Xiao, T. Xu, Y. Xin, and J. Li, "Fbrt-yolo: Faster and better for real-time aerial image detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 8, 2025, pp. 8673–8681.

[5] R. Varghese and S. M., "Yolov8: A novel object detection algorithm with enhanced performance and robustness," in *2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)*, 2024, pp. 1–6.

[6] M. Tan, R. Pang, and Q. V. Le, "Efficientdet: Scalable and efficient object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 10 781–10 790.

[7] W. Liu, H. Lu, H. Fu, and Z. Cao, "Learning to upsample by learning to sample," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 6027–6037.

[8] T. Zhang, Y. Zhuang, G. Wang, S. Dong, H. Chen, and L. Li, "Multiscale semantic fusion-guided fractal convolutional object detection network for optical remote sensing imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–20, 2021.

[9] G. Wang, Y. Zhuang, H. Chen, X. Liu, T. Zhang, L. Li, S. Dong, and Q. Sang, "Fsod-net: Full-scale object detection from optical remote sensing imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–18, 2021.

[10] J. Xue, D. He, M. Liu, and Q. Shi, "Dual network structure with interweaved global-local feature hierarchy for transformer-based object detection in remote sensing image," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 6856–6866, 2022.

[11] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "Abnet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE transactions on geoscience and remote sensing*, vol. 60, pp. 1–14, 2021.

[12] R. Li, S. Zheng, C. Duan, L. Wang, and C. Zhang, "Land cover classification from remote sensing images based on multi-scale fully convolutional network," *Geo-spatial information science*, vol. 25, no. 2, pp. 278–294, 2022.

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